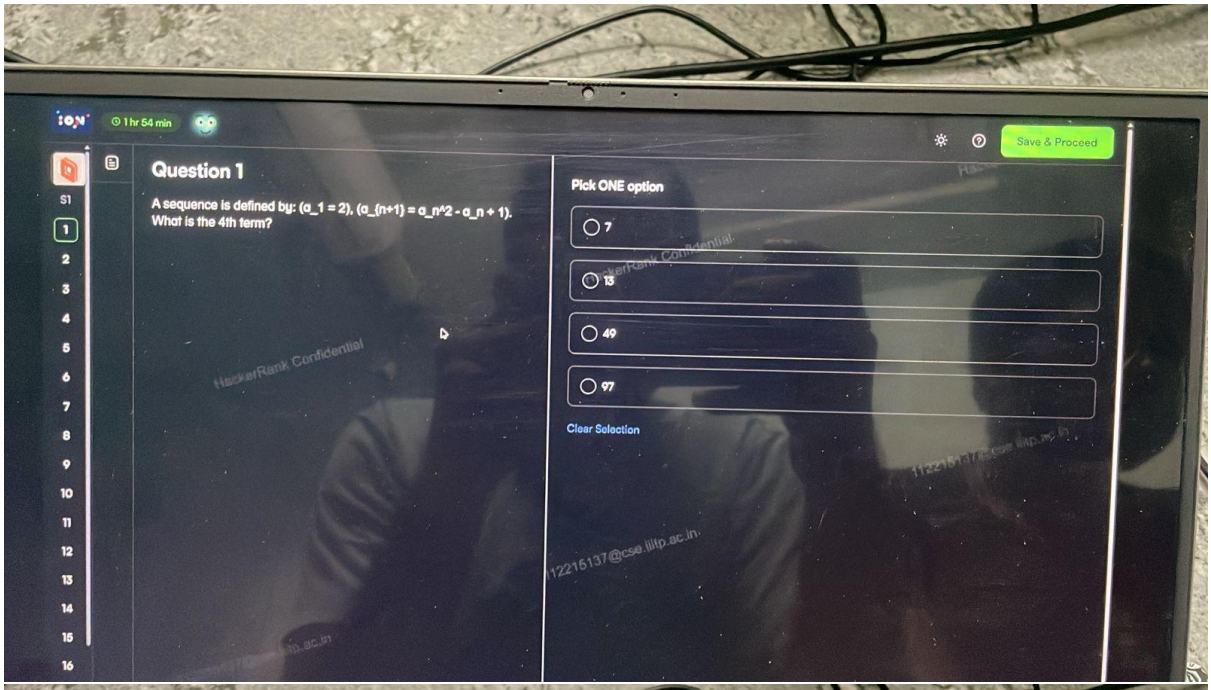
1 hr 54 min
Save & Proceed
Question 1
A sequence is defined by: (a_1 = 2), (a_{n+1} = a_n^2 - a_n + 1).
What is the 4th term?
Pick ONE option
7
13
49
97
Clear Selection
S1
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
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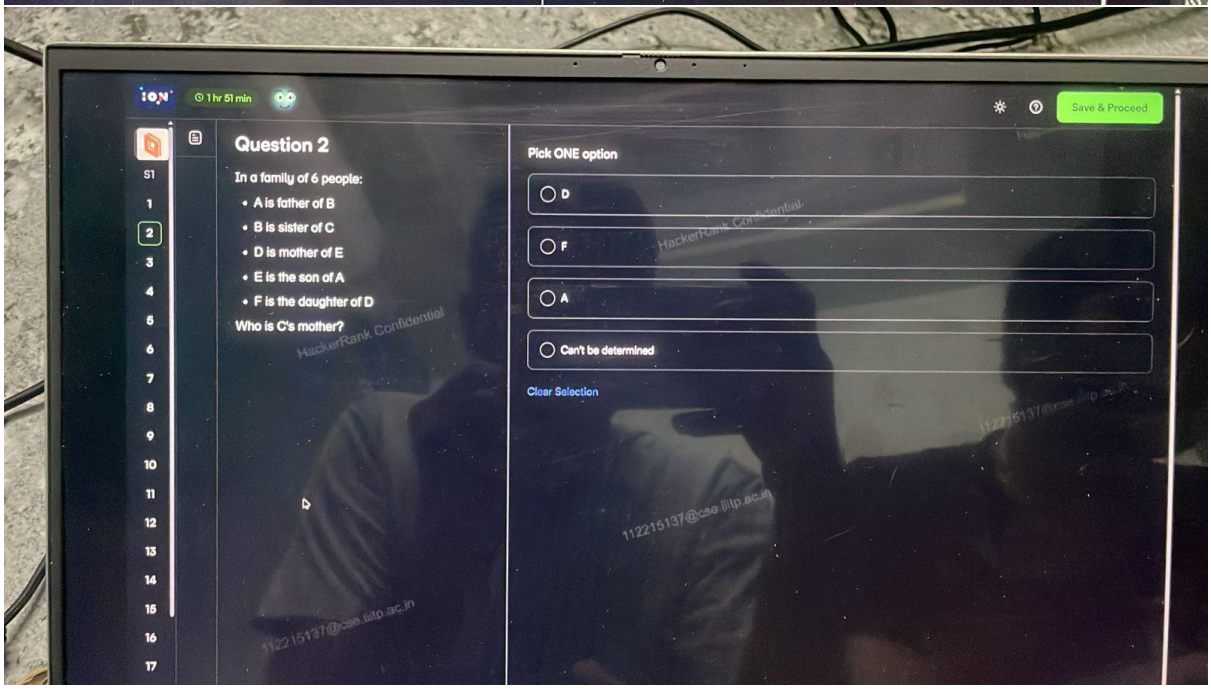
1 hr 51 min
Save & Proceed
Question 2
In a family of 6 people:
A is father of B
B is sister of C
D is mother of E
E is the son of A
F is the daughter of D
Who is C's mother?
Pick ONE option
D
F
A
Can't be determined
Clear Selection
S1
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
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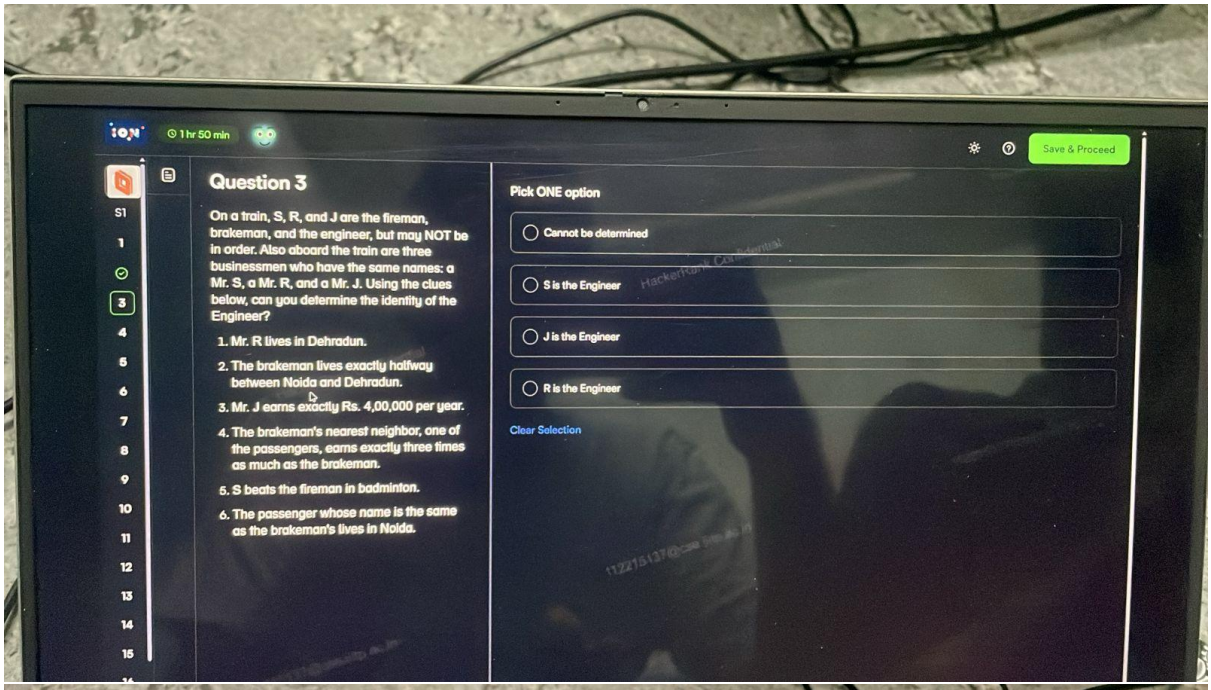
1 hr 50 min
Save & Proceed
S1
1
3
4
5
6
7
8
9
10
11
12
13
14
15
Question 3
On a train, S, R, and J are the fireman, brakeman, and the engineer, but may NOT be in order. Also aboard the train are three businessmen who have the same names: a Mr. S, a Mr. R, and a Mr. J. Using the clues below, can you determine the identity of the Engineer?
1. Mr. R lives in Dehradun.
2. The brakeman lives exactly halfway between Noida and Dehradun.
3. Mr. J earns exactly Rs. 4,00,000 per year.
4. The brakeman's nearest neighbor, one of the passengers, earns exactly three times as much as the brakeman.
5. S beats the fireman in badminton.
6. The passenger whose name is the same as the brakeman's lives in Noida.
Pick ONE option
Cannot be determined
S is the Engineer
J is the Engineer
R is the Engineer
Clear Selection
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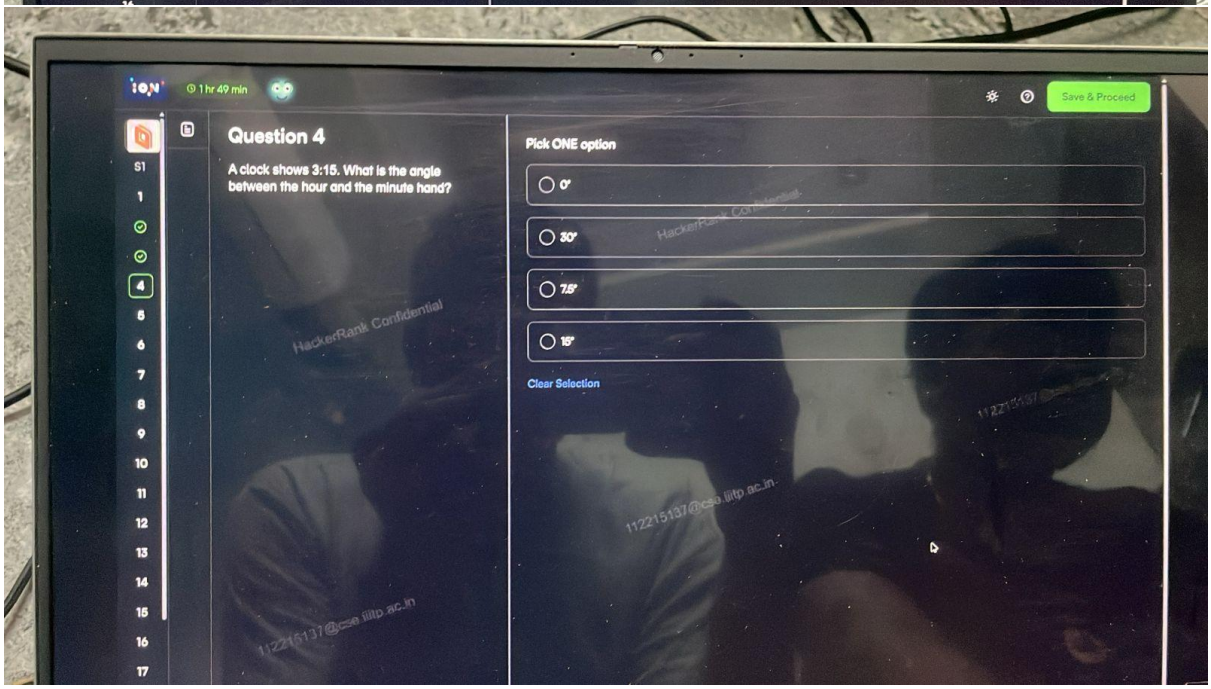
1 hr 49 min
Save & Proceed
S1
1
4
5
6
7
8
9
10
11
12
13
14
15
16
17
Question 4
A clock shows 3:15. What is the angle between the hour and the minute hand?
Pick ONE option
0°
30°
7.5°
15°
Clear Selection
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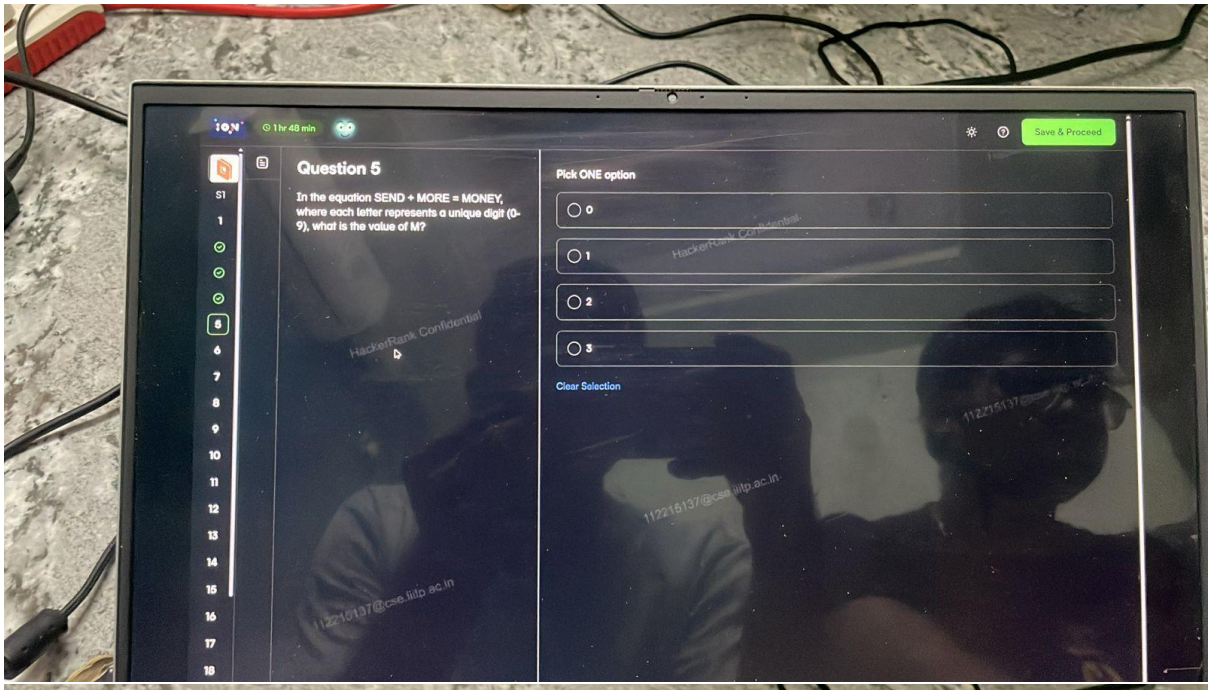
1 hr 48 min
Save & Proceed
Question 5
In the equation SEND + MORE = MONEY, where each letter represents a unique digit (0-9), what is the value of M?
Pick ONE option
0
1
2
3
Clear Selection
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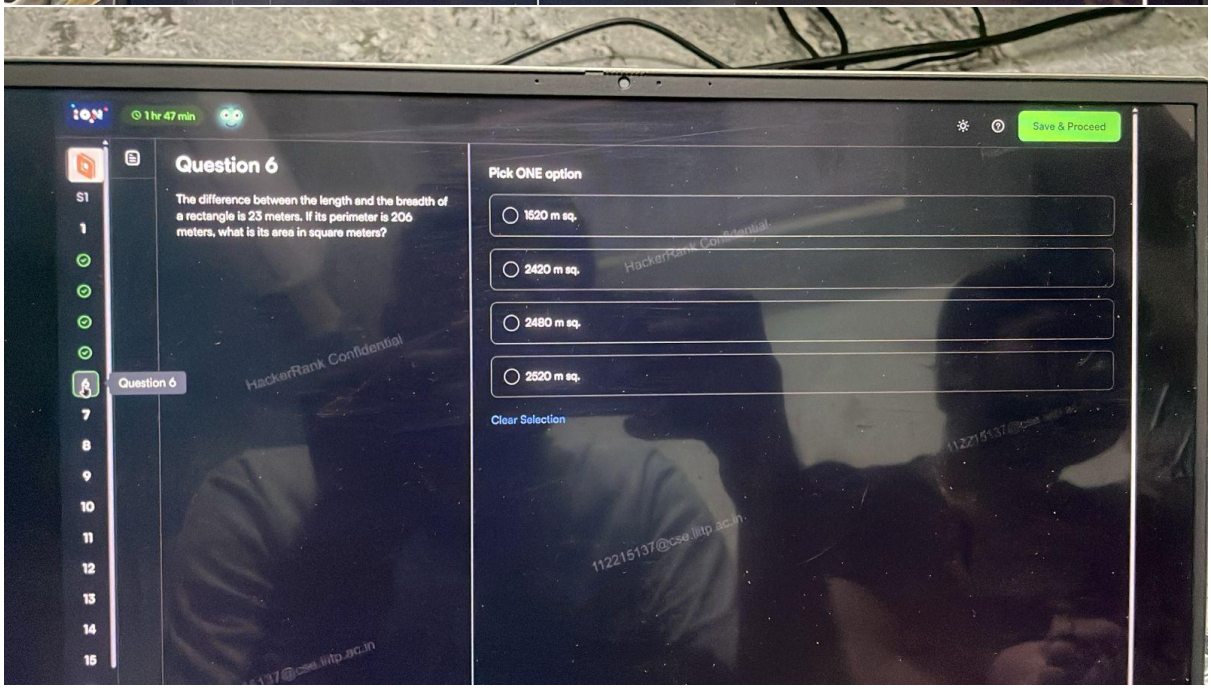
1 hr 47 min
Save & Proceed
Question 6
The difference between the length and the breadth of a rectangle is 23 meters. If its perimeter is 206 meters, what is its area in square meters?
Pick ONE option
1620 m sq.
2420 m sq.
2480 m sq.
2620 m sq.
Clear Selection
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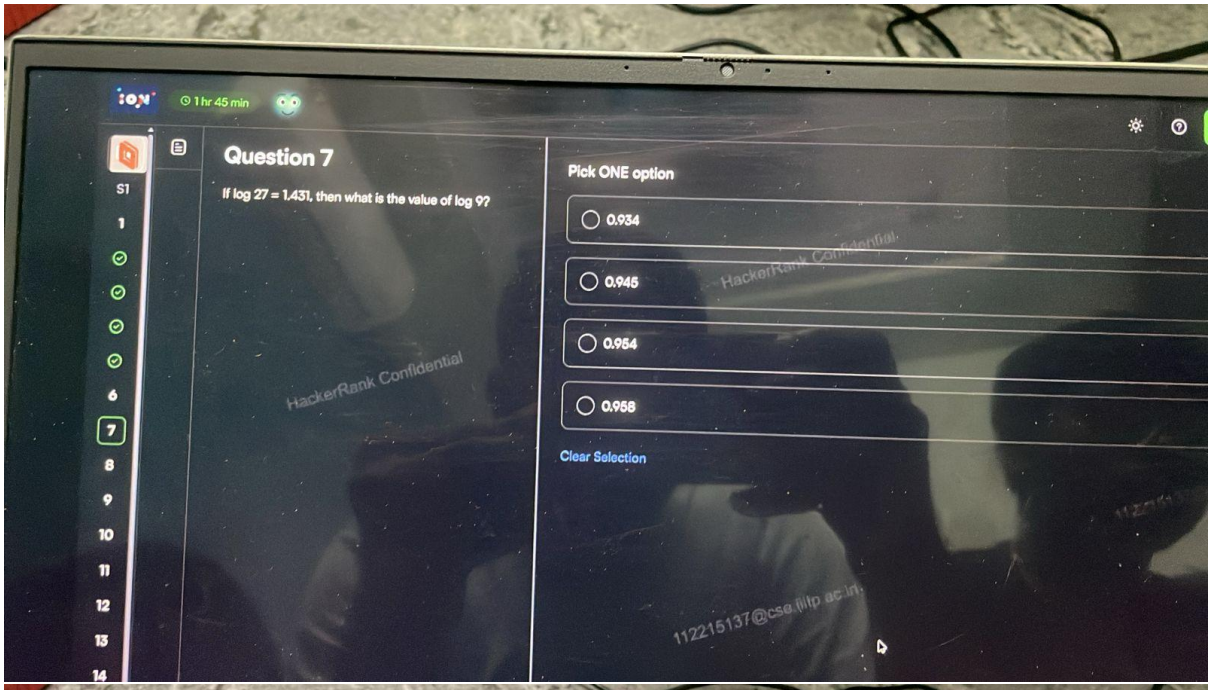
1 hr 45 min
Question 7
If log 27 = 1.431, then what is the value of log 9?
Pick ONE option
0.934
0.945
0.954
0.958
Clear Selection
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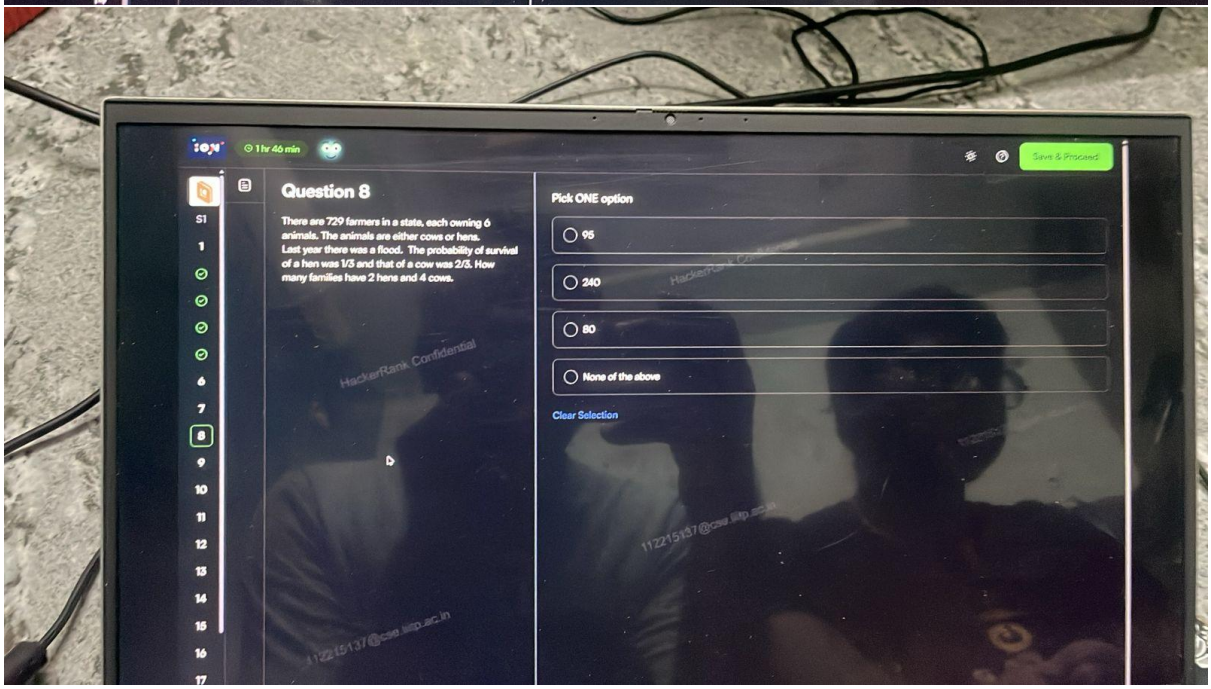
1 hr 46 min
Save & Proceed
Question 8
There are 729 farmers in a state, each owning 6 animals. The animals are either cows or hens. Last year there was a flood. The probability of survival of a hen was 1/3 and that of a cow was 2/3. How many families have 2 hens and 4 cows.
Pick ONE option
95
240
80
None of the above
Clear Selection
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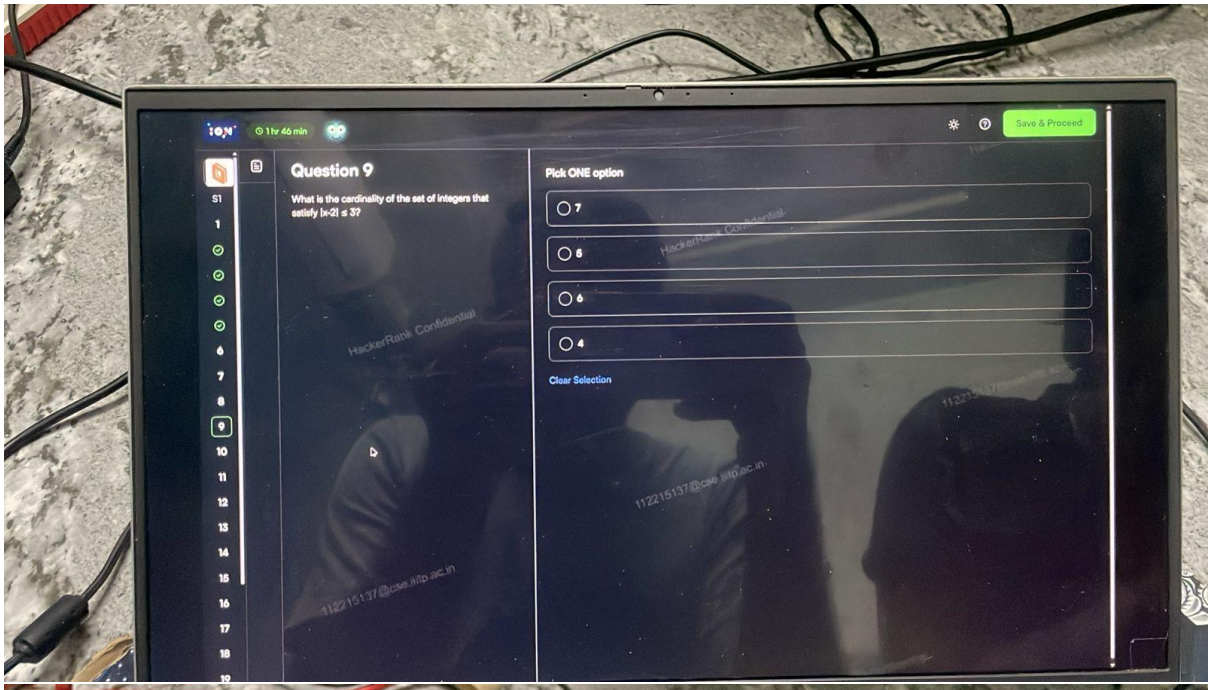
1 hr 46 min
Save & Proceed
Question 9
What is the cardinality of the set of integers that satisfy |x-2| ≤ 3?
Pick ONE option
7
6
4
Clear Selection
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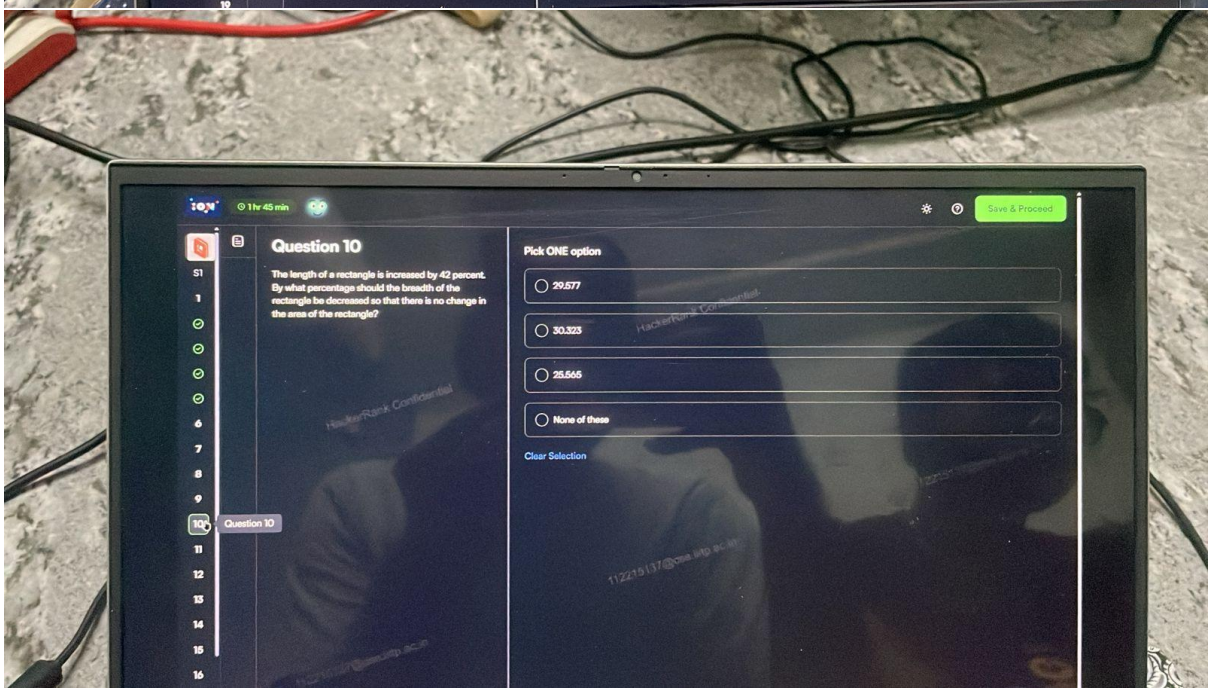
1 hr 45 min
Save & Proceed
Question 10
The length of a rectangle is increased by 42 percent. By what percentage should the breadth of the rectangle be decreased so that there is no change in the area of the rectangle?
Pick ONE option
29.577
30.323
25.565
None of these
Clear Selection
Question 10
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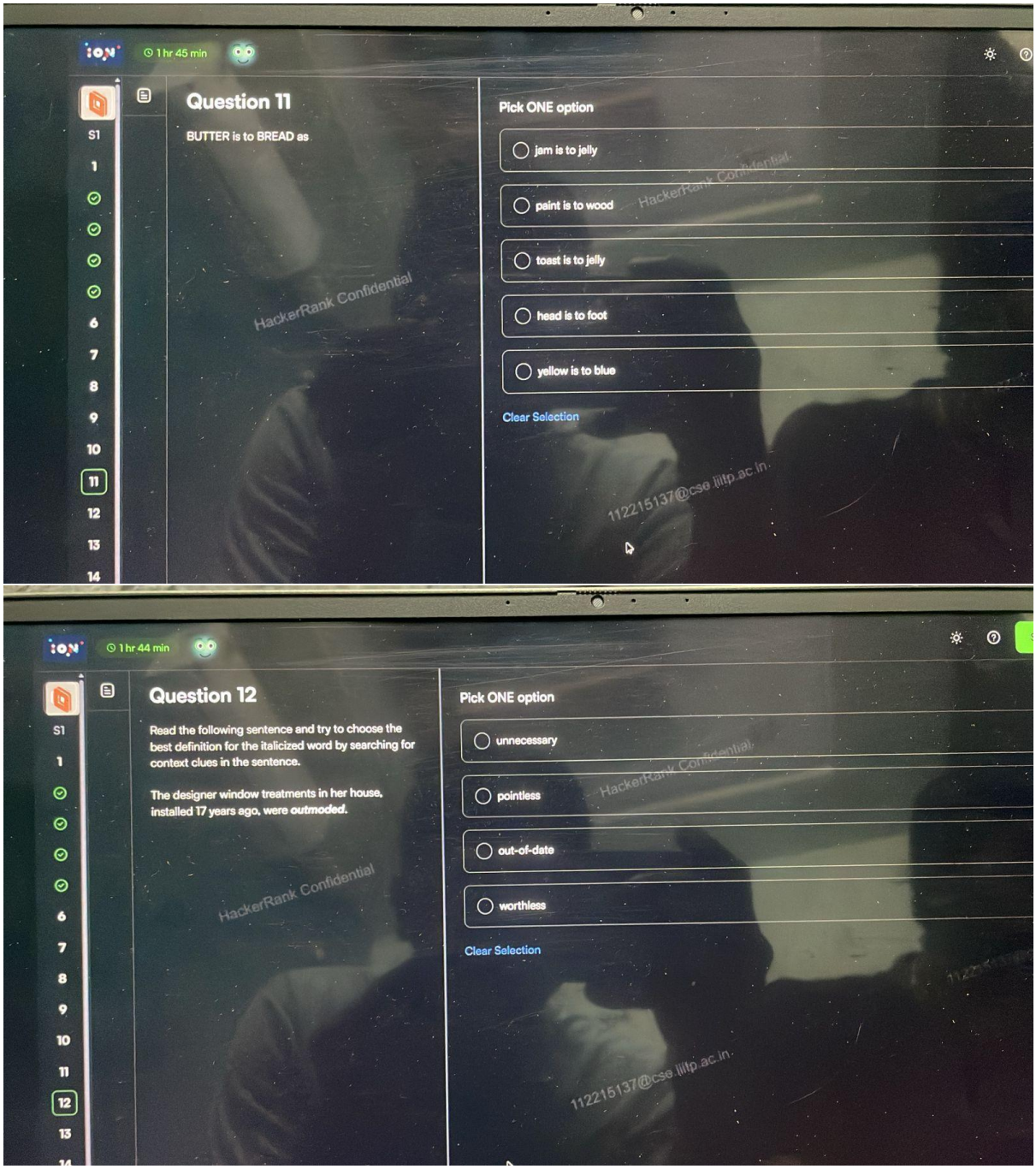
1 hr 45 min
Question 11
BUTTER is to BREAD as
Pick ONE option
jam is to jelly
paint is to wood
toast is to jelly
head is to foot
yellow is to blue
Clear Selection
HackerRank Confidential
112215137@cse.lilp.ac.in
1 hr 44 min
Question 12
Read the following sentence and try to choose the best definition for the italicized word by searching for context clues in the sentence.
The designer window treatments in her house, installed 17 years ago, were outmoded.
Pick ONE option
unnecessary
pointless
out-of-date
worthless
Clear Selection
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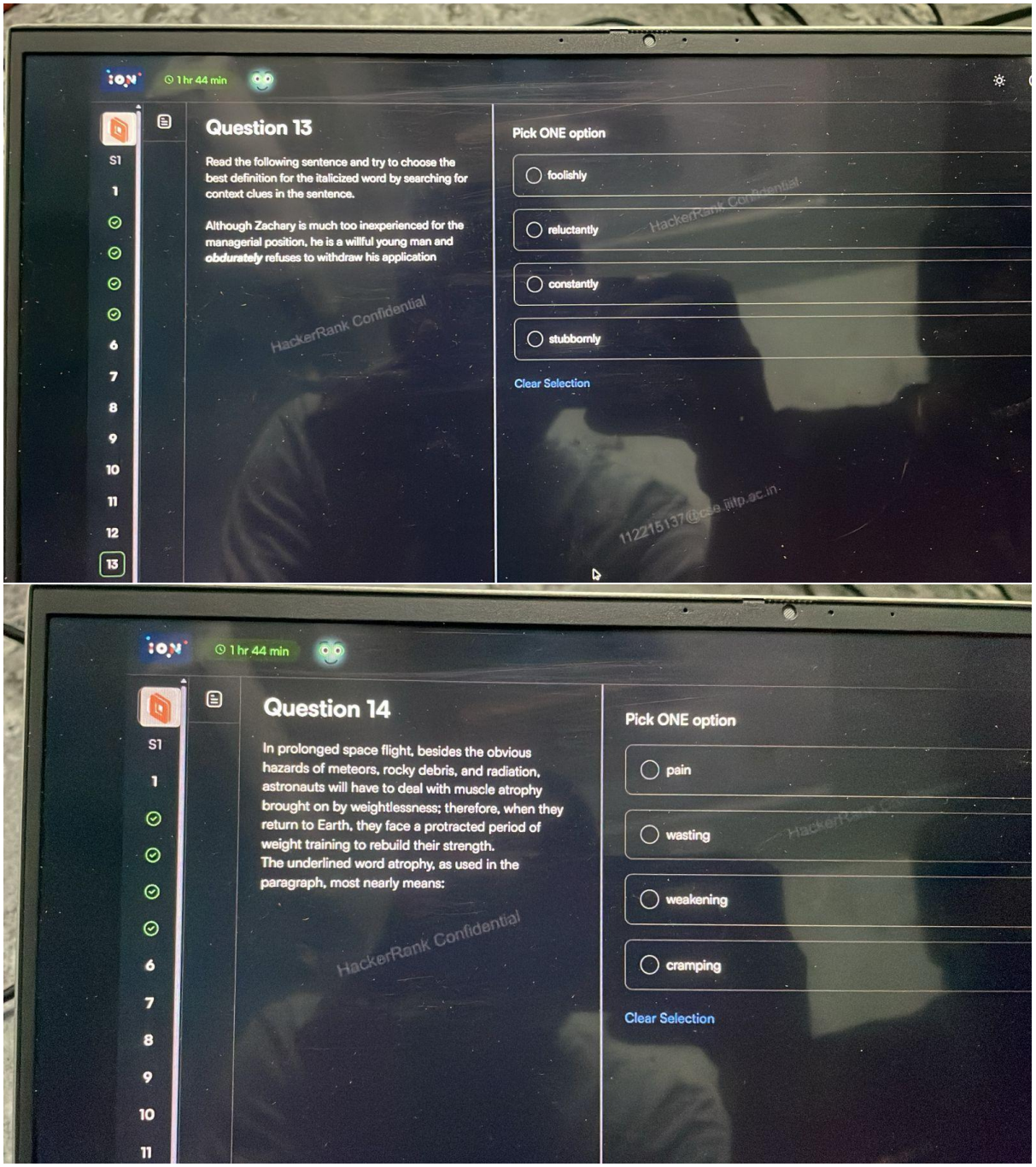
112215137@cse.lilp.ac.in


1 hr 44 min
Question 13
Read the following sentence and try to choose the best definition for the italicized word by searching for context clues in the sentence.
Although Zachary is much too inexperienced for the managerial position, he is a willful young man and obdurately refuses to withdraw his application
HackerRank Confidential
Pick ONE option
foolishly
reluctantly
constantly
stubbornly
Clear Selection
S1
1
6
7
8
9
10
11
12
13
1 hr 44 min
Question 14
In prolonged space flight, besides the obvious hazards of meteors, rocky debris, and radiation, astronauts will have to deal with muscle atrophy brought on by weightlessness; therefore, when they return to Earth, they face a protracted period of weight training to rebuild their strength.
The underlined word atrophy, as used in the paragraph, most nearly means:
HackerRank Confidential
Pick ONE option
pain
wasting
weakening
cramping
Clear Selection
S1
1
6
7
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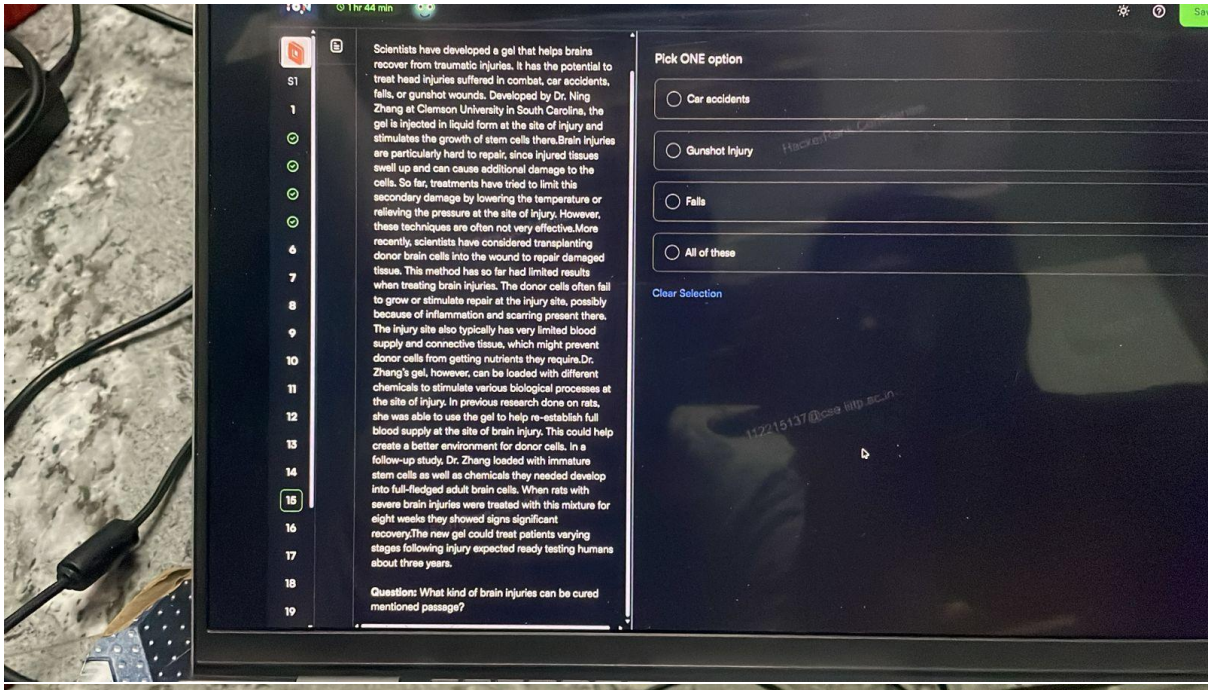
9
10
11


1 hr 44 min
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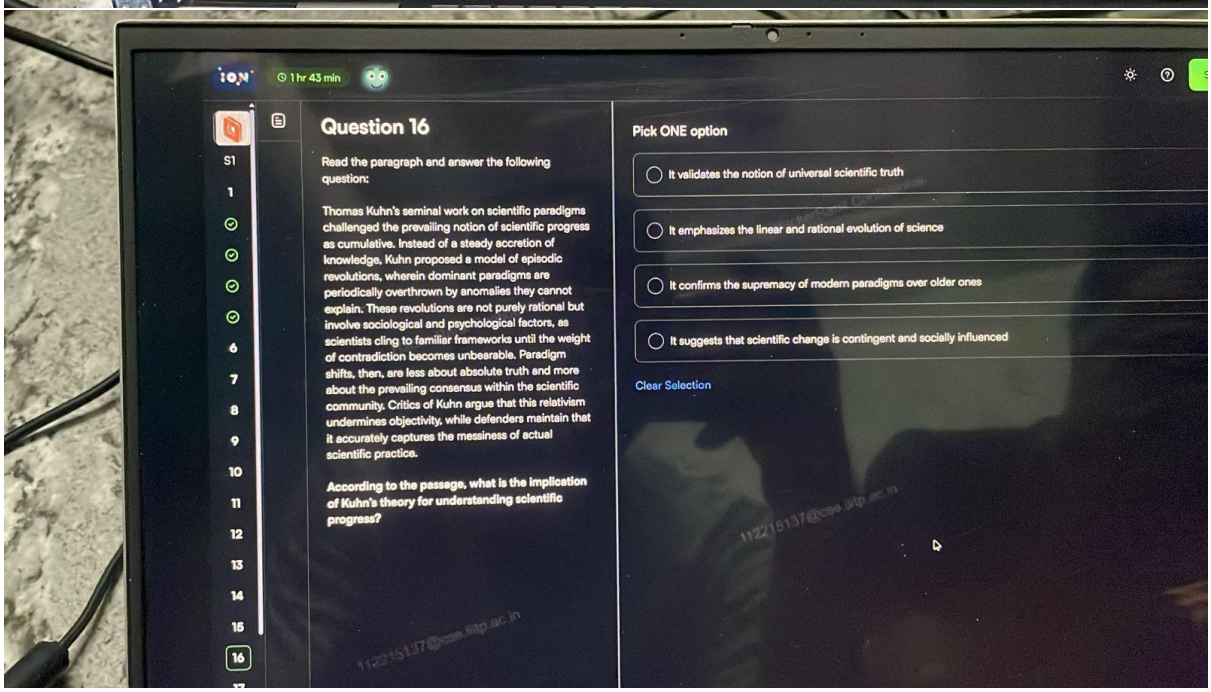
19
Scientists have developed a gel that helps brains recover from traumatic injuries. It has the potential to treat head injuries suffered in combat, car accidents, falls, or gunshot wounds. Developed by Dr. Ning Zhang at Clemson University in South Carolina, the gel is injected in liquid form at the site of injury and stimulates the growth of stem cells there.Brain injuries are particularly hard to repair, since injured tissues swell up and can cause additional damage to the cells. So far, treatments have tried to limit this secondary damage by lowering the temperature or relieving the pressure at the site of injury. However, these techniques are often not very effective.More recently, scientists have considered transplanting donor brain cells into the wound to repair damaged tissue. This method has so far had limited results when treating brain injuries. The donor cells often fail to grow or stimulate repair at the injury site, possibly because of inflammation and scarring present there. The injury site also typically has very limited blood supply and connective tissue, which might prevent donor cells from getting nutrients they require.Dr. Zhang's gel, however, can be loaded with different chemicals to stimulate various biological processes at the site of injury. In previous research done on rats, she was able to use the gel to help re-establish full blood supply at the site of brain injury. This could help create a better environment for donor cells. In a follow-up study, Dr. Zhang loaded with immature stem cells as well as chemicals they needed develop into full-fledged adult brain cells. When rats with severe brain injuries were treated with this mixture for eight weeks they showed signs significant recovery.The new gel could treat patients varying stages following injury expected ready testing humans about three years.
Question: What kind of brain injuries can be cured mentioned passage?
Pick ONE option
Car accidents
Gunshot Injury
Falls
All of these
Clear Selection


1 hr 43 min
S1
1
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11
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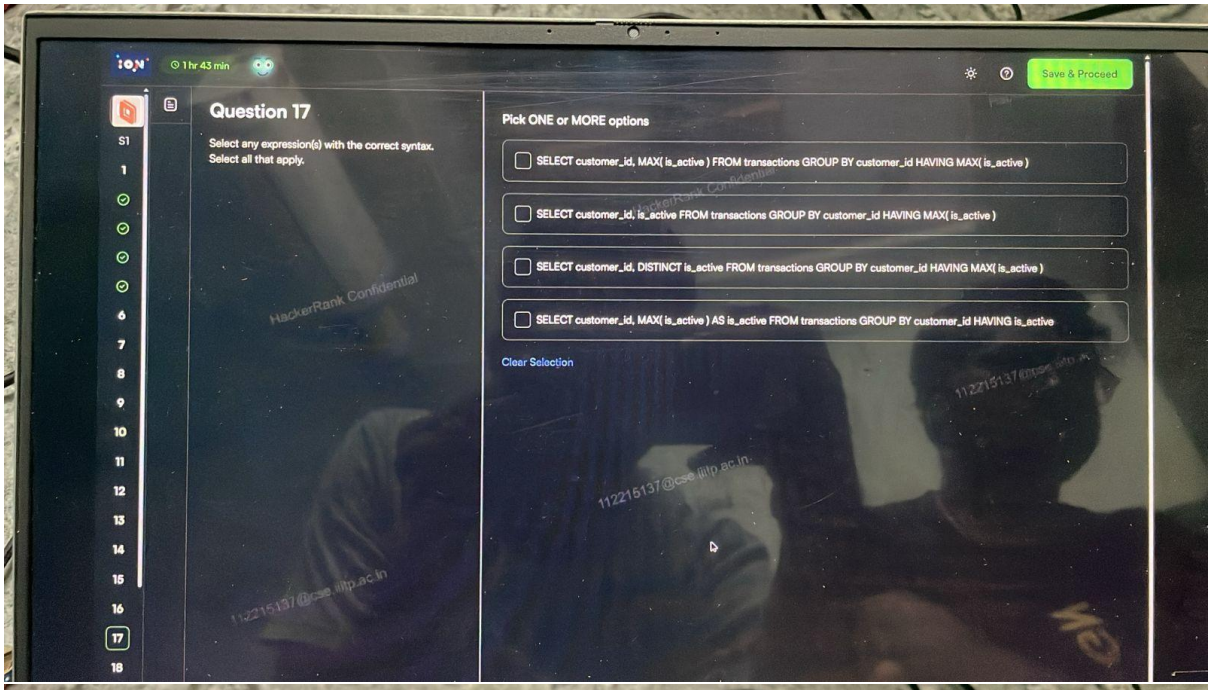
14
15
16
Question 16
Read the paragraph and answer the following question:
Thomas Kuhn's seminal work on scientific paradigms challenged the prevailing notion of scientific progress as cumulative. Instead of a steady accretion of knowledge, Kuhn proposed a model of episodic revolutions, wherein dominant paradigms are periodically overthrown by anomalies they cannot explain. These revolutions are not purely rational but involve sociological and psychological factors, as scientists cling to familiar frameworks until the weight of contradiction becomes unbearable. Paradigm shifts, then, are less about absolute truth and more about the prevailing consensus within the scientific community. Critics of Kuhn argue that this relativism undermines objectivity, while defenders maintain that it accurately captures the messiness of actual scientific practice.
According to the passage, what is the implication of Kuhn's theory for understanding scientific progress?
Pick ONE option
It validates the notion of universal scientific truth
It emphasizes the linear and rational evolution of science
It confirms the supremacy of modern paradigms over older ones
It suggests that scientific change is contingent and socially influenced
Clear Selection


1 hr 43 min
Save & Proceed
Question 17
Select any expression(s) with the correct syntax. Select all that apply.
Pick ONE or MORE options
SELECT customer_id, MAX(is_active) FROM transactions GROUP BY customer_id HAVING MAX(is_active)
SELECT customer_id, is_active FROM transactions GROUP BY customer_id HAVING MAX(is_active)
SELECT customer_id, DISTINCT is_active FROM transactions GROUP BY customer_id HAVING MAX(is_active)
SELECT customer_id, MAX(is_active) AS is_active FROM transactions GROUP BY customer_id HAVING is_active
Clear Selection
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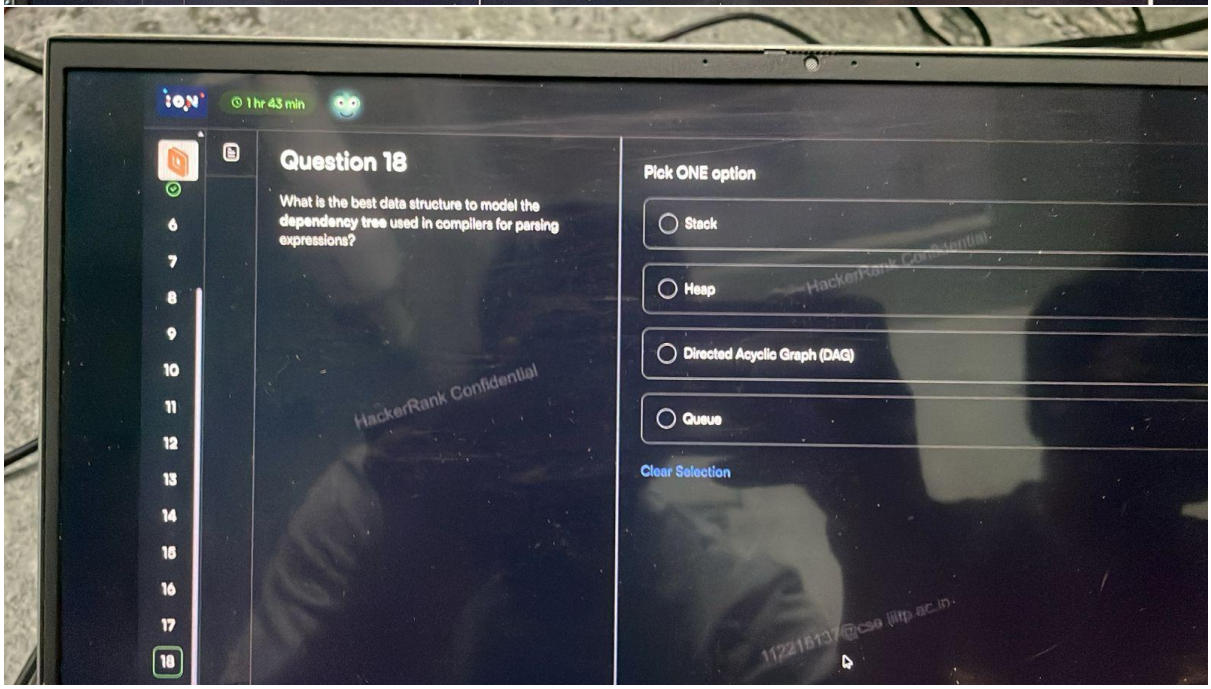
1 hr 43 min
Question 18
What is the best data structure to model the dependency tree used in compilers for parsing expressions?
Pick ONE option
Stack
Heap
Directed Acyclic Graph (DAG)
Queue
Clear Selection
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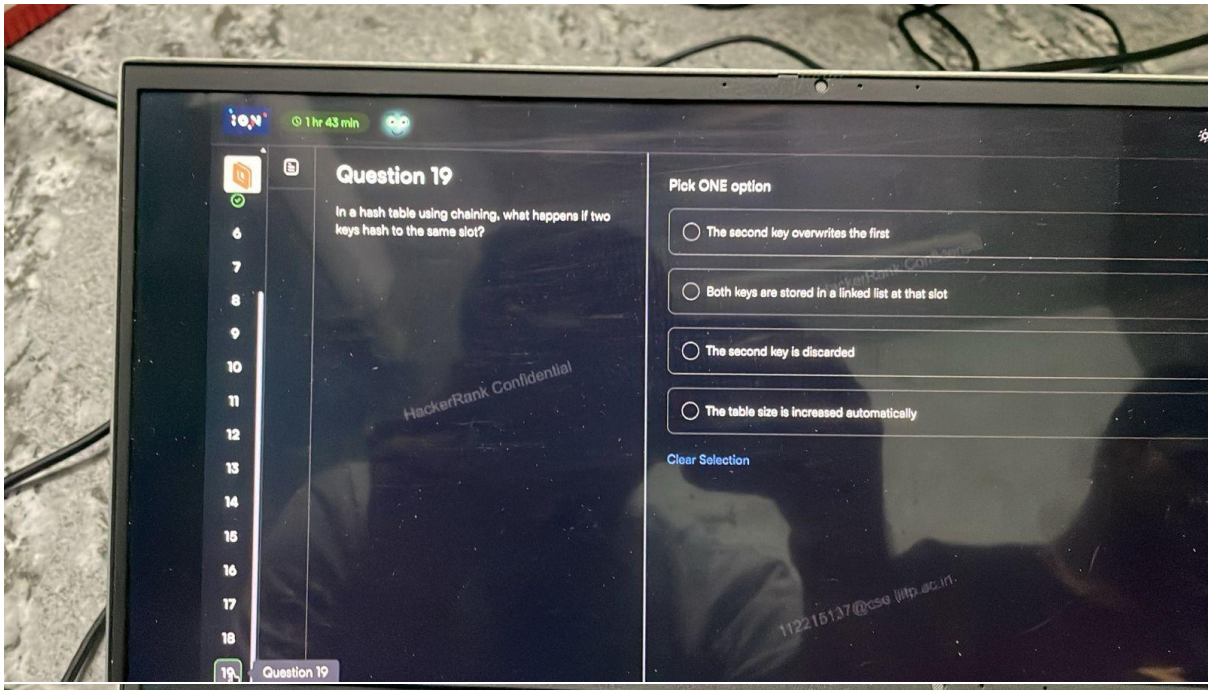
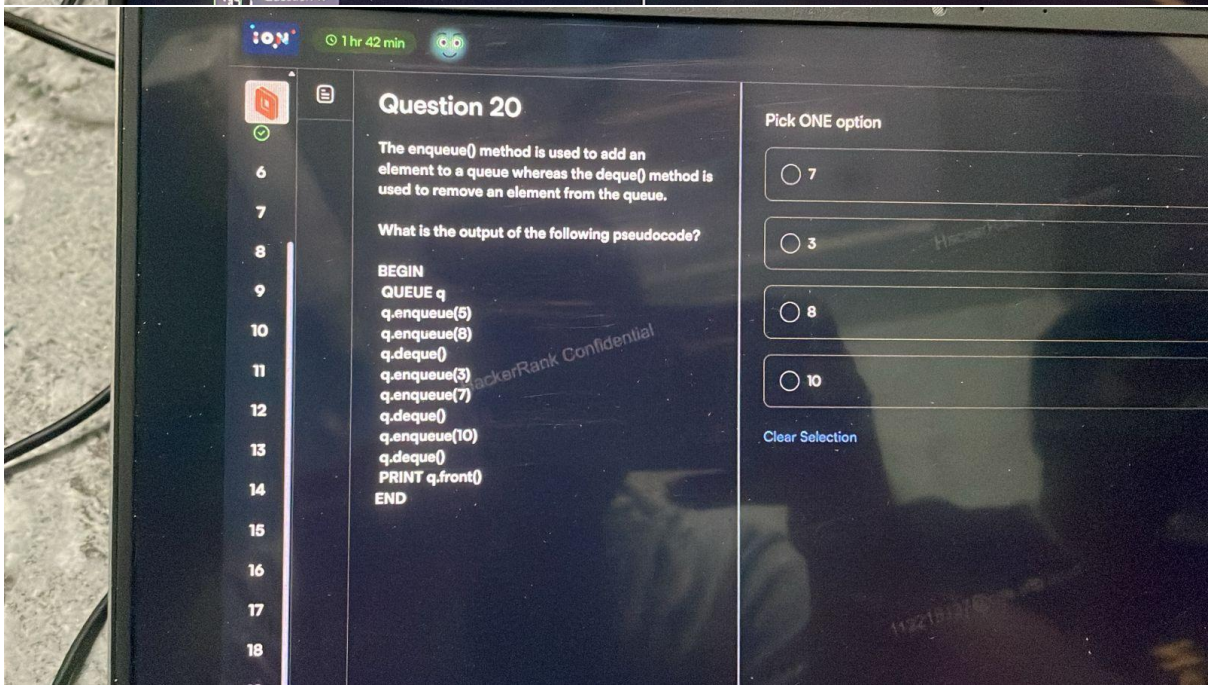
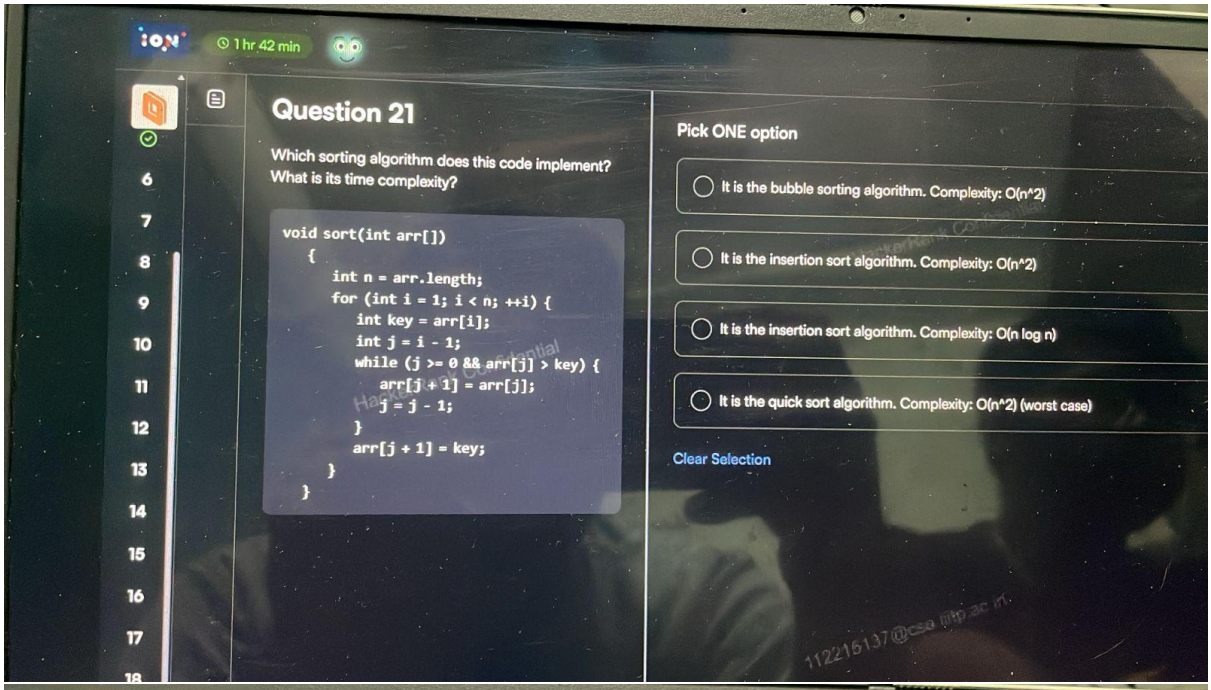
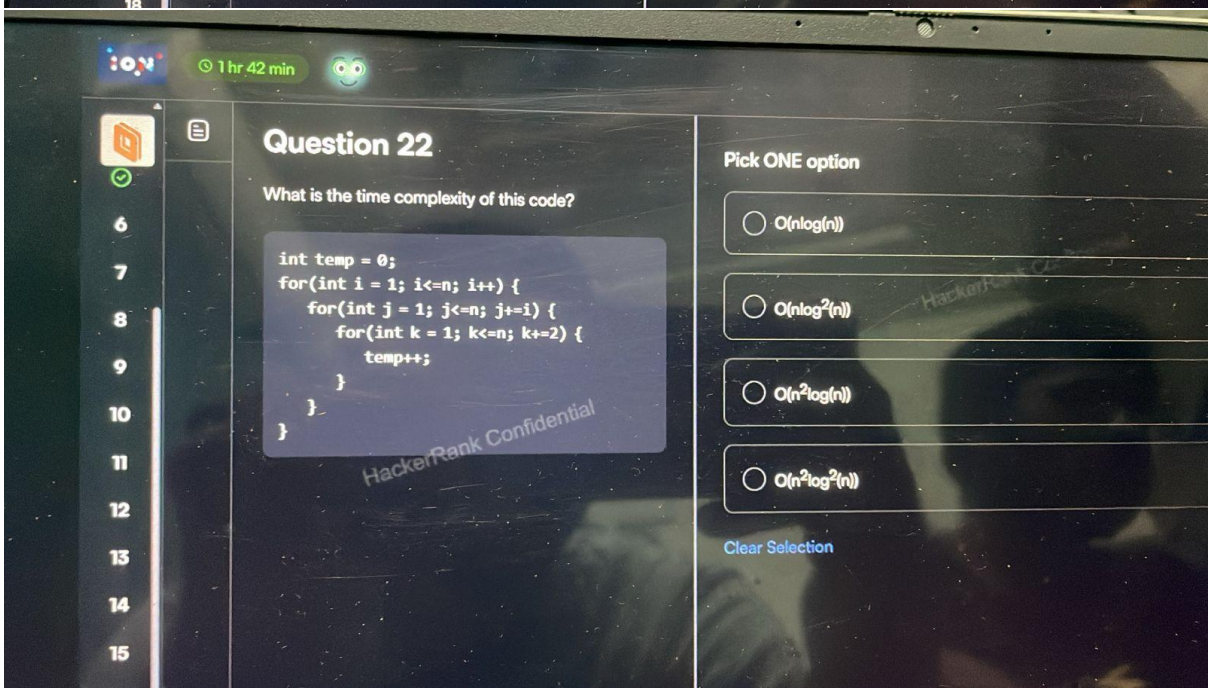
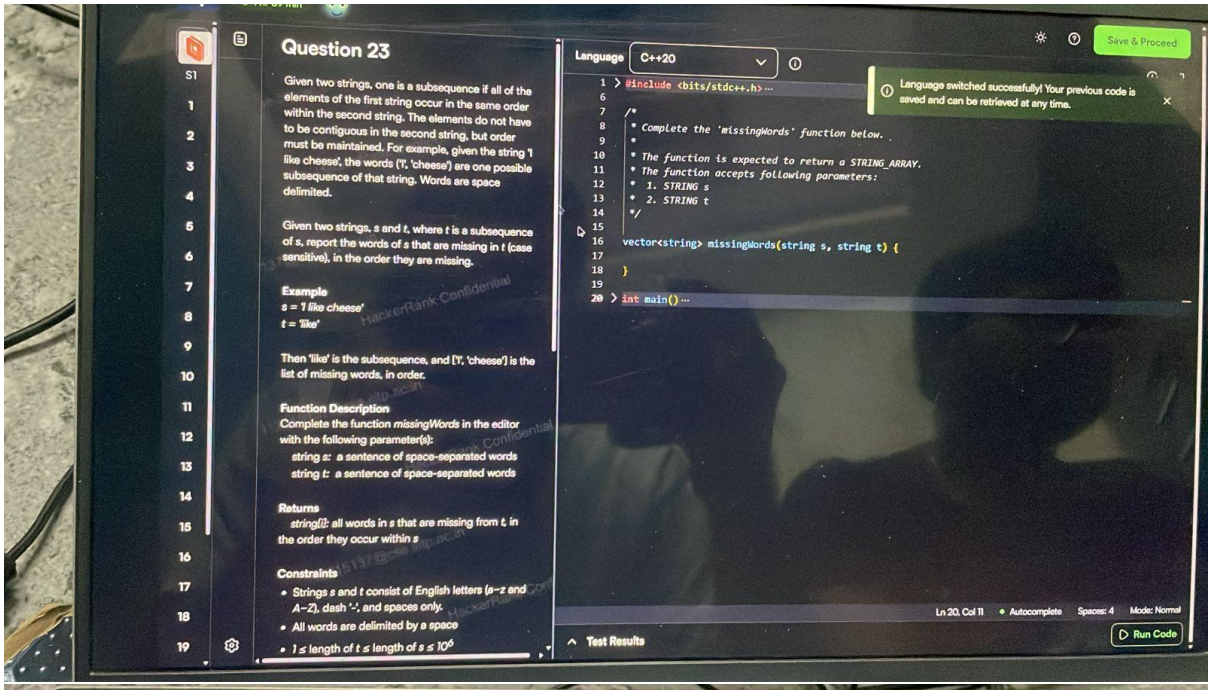
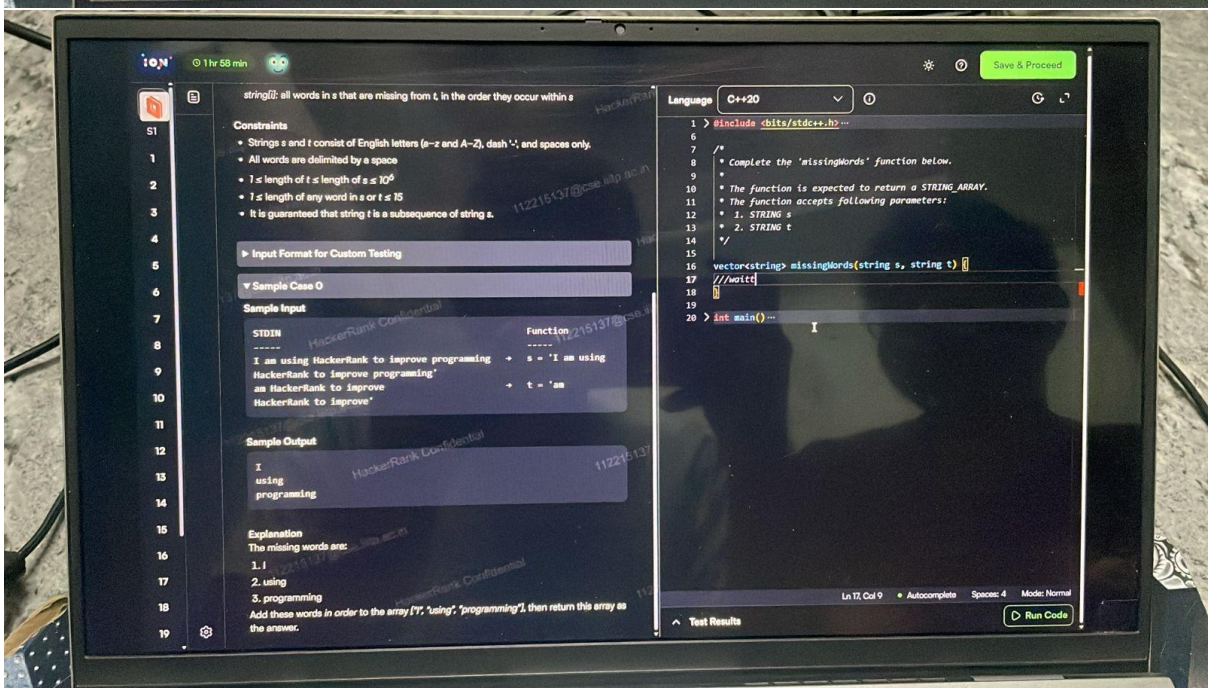

1 hr 43 min
Question 19
In a hash table using chaining, what happens if two keys hash to the same slot?
Pick ONE option
The second key overwrites the first
Both keys are stored in a linked list at that slot
The second key is discarded
The table size is increased automatically
Clear Selection
HackerRank Confidential
Question 19


1 hr 42 min
Question 20
The enqueue() method is used to add an element to a queue whereas the deque() method is used to remove an element from the queue.
What is the output of the following pseudocode?
BEGIN
QUEUE q
q.enqueue(5)
q.enqueue(8)
q.deque()
q.enqueue(3)
q.enqueue(7)
q.deque()
q.enqueue(10)
q.deque()
PRINT q.front()
END
Pick ONE option
7
3
8
10
Clear Selection
HackerRank Confidential

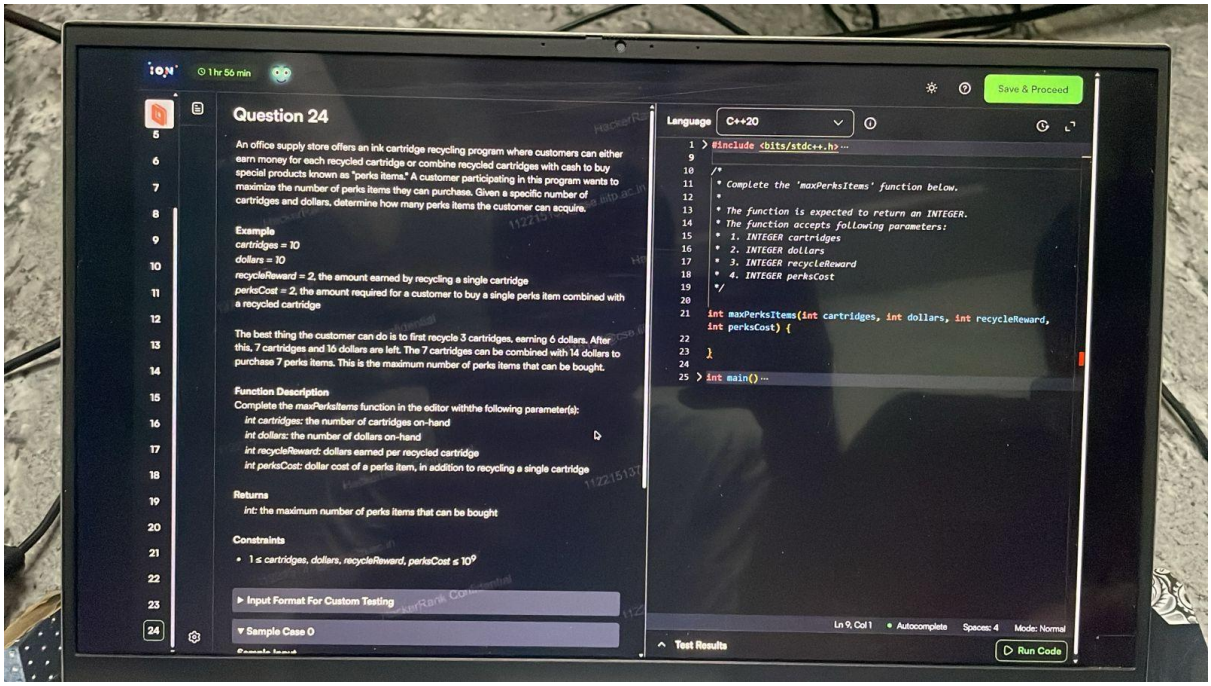

1 hr 42 min
Question 21
Which sorting algorithm does this code implement? What is its time complexity?
void sort(int arr[])
{
int n = arr.length;
for (int i = 1; i < n; ++i) {
int key = arr[i];
int j = i - 1;
while (j >= 0 && arr[j] > key) {
arr[j + 1] = arr[j];
j = j - 1;
}
arr[j + 1] = key;
}
}
Pick ONE option
It is the bubble sorting algorithm. Complexity: O(n^2)
It is the insertion sort algorithm. Complexity: O(n^2)
It is the insertion sort algorithm. Complexity: O(n log n)
It is the quick sort algorithm. Complexity: O(n^2) (worst case)
Clear Selection

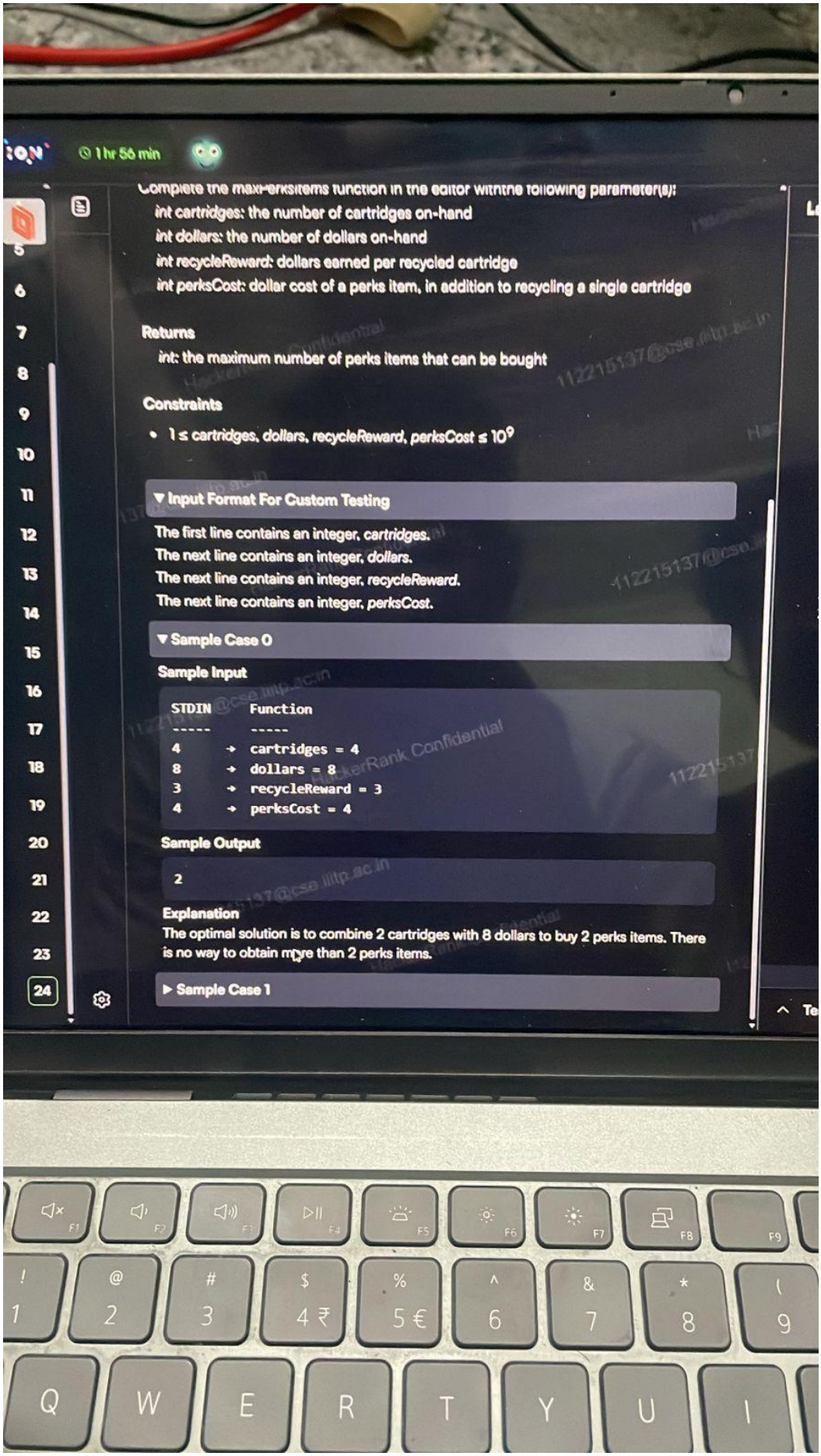

1 hr 42 min
Question 22
What is the time complexity of this code?
int temp = 0;
for(int i = 1; i<=n; i++) {
for(int j = 1; j<=n; j+=i) {
for(int k = 1; k<=n; k+=2) {
temp++;
}
}
}
HackerRank Confidential
Pick ONE option
O(nlog(n))
O(nlog^2(n))
O(n^{2}log(n))
O(n^{2}log^2(n))
Clear Selection

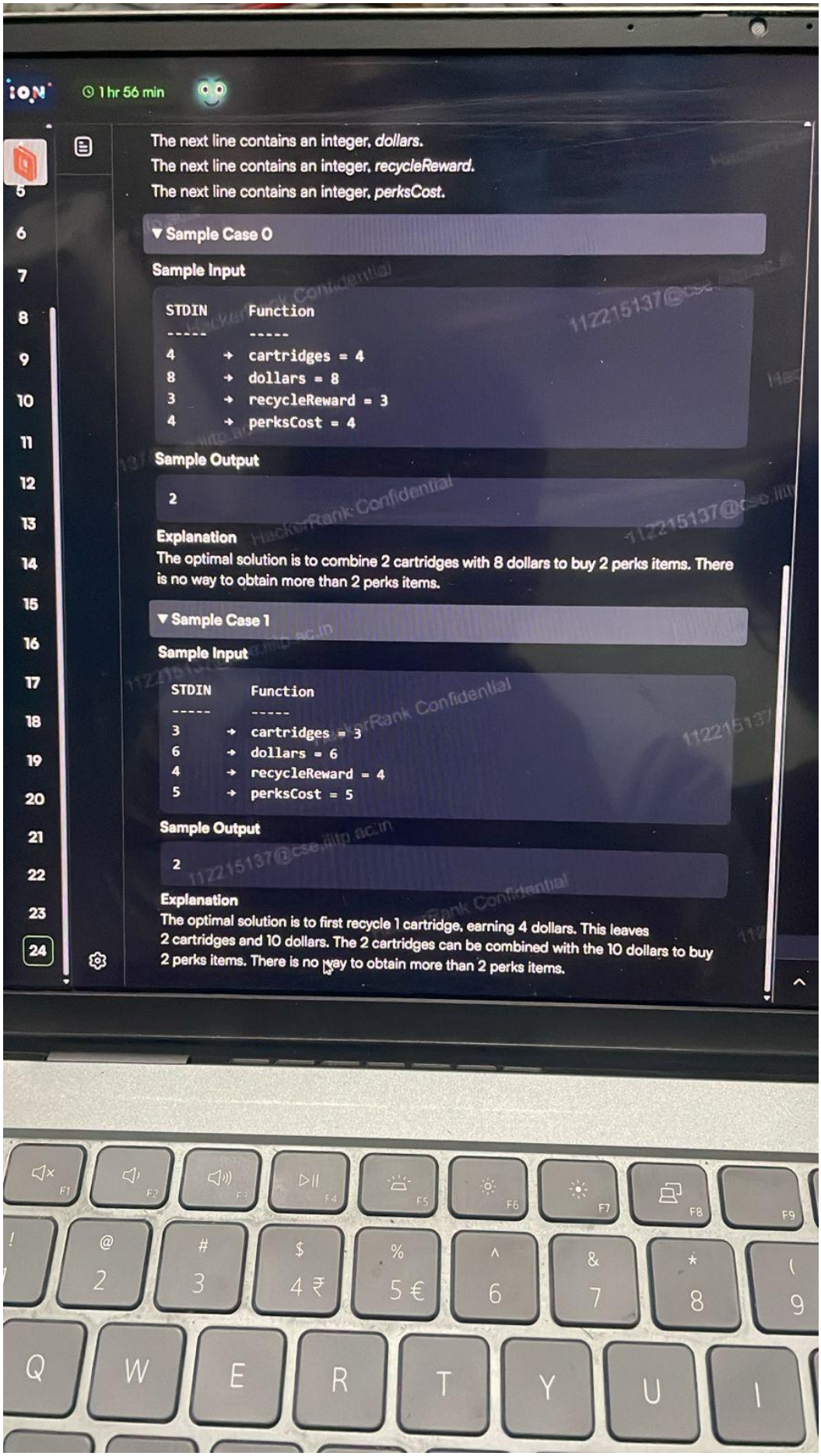

Question 23
Given two strings, one is a subsequence if all of the elements of the first string occur in the same order within the second string. The elements do not have to be contiguous in the second string, but order must be maintained. For example, given the string 'I like cheese', the words ('I', 'cheese') are one possible subsequence of that string. Words are space delimited.
Given two strings, s and t, where t is a subsequence of s, report the words of s that are missing in t (case sensitive), in the order they are missing.
Example
s = 'I like cheese'
t = 'like'
Then 'like' is the subsequence, and ['I', 'cheese'] is the list of missing words, in order.
Function Description
Complete the function missingWords in the editor with the following parameter(s):
string s: a sentence of space-separated words
string t: a sentence of space-separated words
Returns
string[i]: all words in s that are missing from t, in the order they occur within s
Constraints
Strings s and t consist of English letters (a–z and A–Z), dash '-', and spaces only.
All words are delimited by a space
1 ≤ length of t ≤ length of s ≤ 10^6
Language
C++20
Save & Proceed
Language switched successfully! Your previous code is saved and can be retrieved at any time.
#include <bits/stdc++.h>
/*
* Complete the 'missingWords' function below.
*
* The function is expected to return a STRING_ARRAY.
* The function accepts following parameters:
* 1. STRING s
* 2. STRING t
*/
vector<string> missingWords(string s, string t) {
}
int main()
Ln 20, Col 11
Autocomplete
Spaces: 4
Mode: Normal
Test Results
Run Code


1 hr 58 min
Save & Proceed
string[i]: all words in s that are missing from t, in the order they occur within s
Constraints
Strings s and t consist of English letters (a–z and A–Z), dash '-', and spaces only.
All words are delimited by a space
1 ≤ length of t ≤ length of s ≤ 10^6
1 ≤ length of any word in s or t ≤ 15
It is guaranteed that string t is a subsequence of string s.
Input Format for Custom Testing
Sample Case 0
Sample Input
STDIN
Function
I am using HackerRank to improve programming
HackerRank to improve programming'
am HackerRank to improve
HackerRank to improve'
s = 'I am using
t = 'am
Sample Output
I
using
programming
Explanation
The missing words are:
1. I
2. using
3. programming
Add these words in order to the array ['I', 'using', 'programming'], then return this array as the answer.
Language
C++20
#include <bits/stdc++.h>
/*
* Complete the 'missingWords' function below.
*
* The function is expected to return a STRING_ARRAY.
* The function accepts following parameters:
* 1. STRING s
* 2. STRING t
*/
vector<string> missingWords(string s, string t) {
///write
}
int main()
Ln 17, Col 9
Autocomplete
Spaces: 4
Mode: Normal
Test Results
Run Code

```cpp
vector<string> missingWords(string s, string t) {
    vector<string> result;
    istringstream ss(s), tt(t);
    vector<string> s_words, t_words;

    string word;
    while (ss >> word) s_words.push_back(word);
    while (tt >> word) t_words.push_back(word);

    int i = 0, j = 0;
    while (i < s_words.size()) {
        if (j < t_words.size() && s_words[i] == t_words[j]) {
            j++;
        } else {
            result.push_back(s_words[i]);
        }
        i++;
    }

    return result;
}
```




1 hr 56 min
Complete the maxiPerksItems function in the editor withthe following parameter(s):
int cartridges: the number of cartridges on-hand
int dollars: the number of dollars on-hand
int recycleReward: dollars earned per recycled cartridge
int perksCost: dollar cost of a perks item, in addition to recycling a single cartridge
Returns
int: the maximum number of perks items that can be bought
Constraints
1 ≤ cartridges, dollars, recycleReward, perksCost ≤ 10^9
Input Format For Custom Testing
The first line contains an integer, cartridges.
The next line contains an integer, dollars.
The next line contains an integer, recycleReward.
The next line contains an integer, perksCost.
Sample Case 0
Sample Input
STDIN Function
----- -----
4 → cartridges = 4
8 → dollars = 8
3 → recycleReward = 3
4 → perksCost = 4
Sample Output
2
Explanation
The optimal solution is to combine 2 cartridges with 8 dollars to buy 2 perks items. There is no way to obtain more than 2 perks items.
Sample Case 1



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The next line contains an integer, *dollars*.
The next line contains an integer, *recycleReward*.
The next line contains an integer, *perksCost*.

▼ Sample Case 0

Sample Input

```
STDIN       Function
-----       -----
4       →   cartridges = 4
8       →   dollars = 8
3       →   recycleReward = 3
4       →   perksCost = 4
```

Sample Output

```
2
```

Explanation
The optimal solution is to combine 2 cartridges with 8 dollars to buy 2 perks items. There is no way to obtain more than 2 perks items.

▼ Sample Case 1

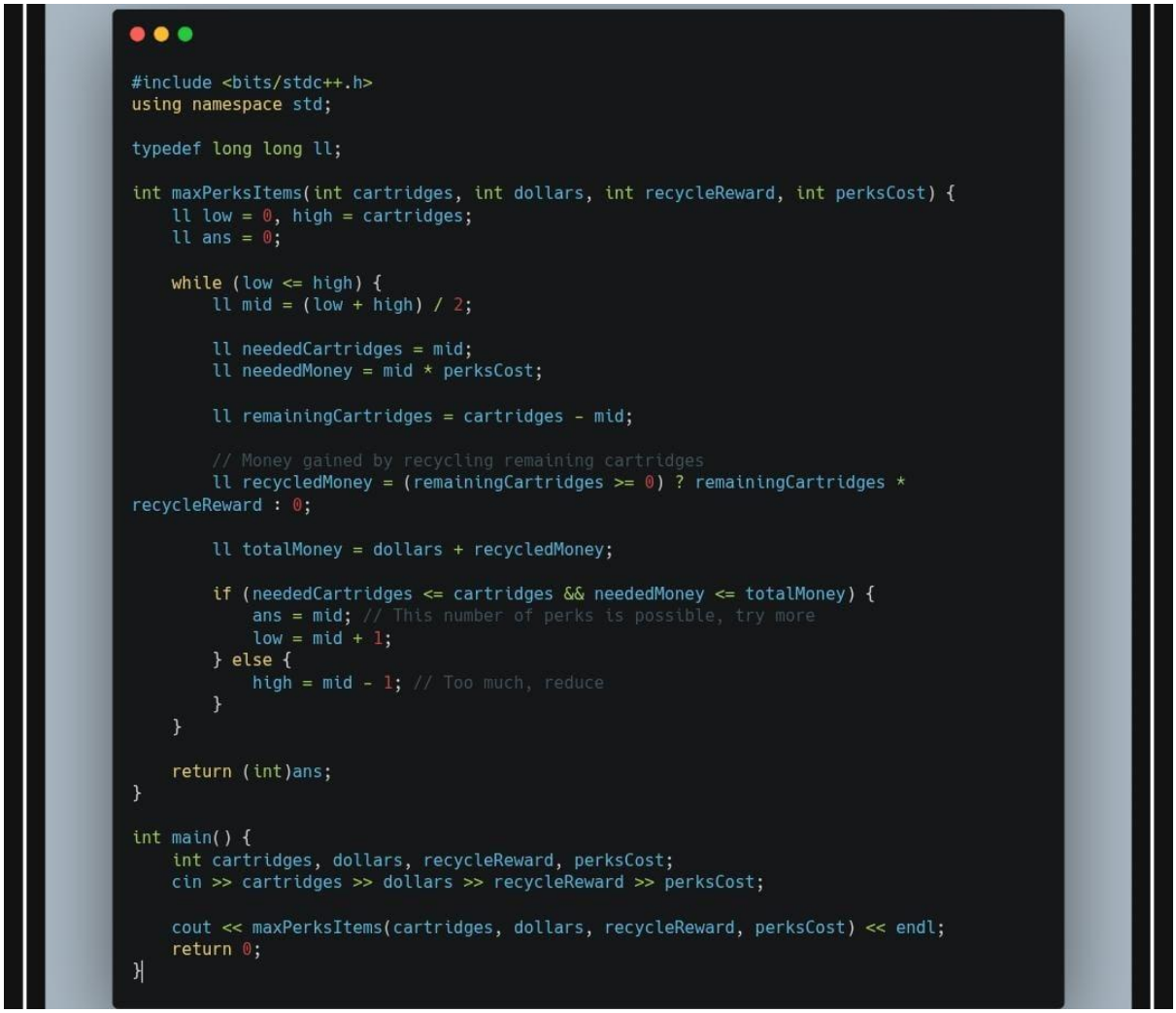
Sample Input

```
STDIN       Function
-----       -----
3       →   cartridges = 3
6       →   dollars = 6
4       →   recycleReward = 4
5       →   perksCost = 5
```

Sample Output

```
2
```

Explanation
The optimal solution is to first recycle 1 cartridge, earning 4 dollars. This leaves 2 cartridges and 10 dollars. The 2 cartridges can be combined with the 10 dollars to buy 2 perks items. There is no way to obtain more than 2 perks items.



```cpp
#include <bits/stdc++.h>
using namespace std;

typedef long long ll;

int maxPerksItems(int cartridges, int dollars, int recycleReward, int perksCost) {
    ll low = 0, high = cartridges;
    ll ans = 0;

    while (low <= high) {
        ll mid = (low + high) / 2;

        ll neededCartridges = mid;
        ll neededMoney = mid * perksCost;

        ll remainingCartridges = cartridges - mid;

        // Money gained by recycling remaining cartridges
        ll recycledMoney = (remainingCartridges >= 0) ? remainingCartridges *
recycleReward : 0;

        ll totalMoney = dollars + recycledMoney;

        if (neededCartridges <= cartridges && neededMoney <= totalMoney) {
            ans = mid; // This number of perks is possible, try more
            low = mid + 1;
        } else {
            high = mid - 1; // Too much, reduce
        }
    }

    return (int)ans;
}

int main() {
    int cartridges, dollars, recycleReward, perksCost;
    cin >> cartridges >> dollars >> recycleReward >> perksCost;

    cout << maxPerksItems(cartridges, dollars, recycleReward, perksCost) << endl;
    return 0;
}
```